PALAEOMAGNETISM OF THE ABITIBI DYKE SWARM

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ABSTRACT

Previous data published on the palaeomagnetism of a group of Precambrian diabase dykes, referred to as the Abitibi swarm, were characterized by considerable angular dispersion attributed to a number of possible factors. The relative importance of these factors was investigated in the light of data obtained for an independent suite of samples from the same group of dykes. Most of the previous interpretation is probably no longer valid because the magnetization of the swarm was found to be distributed tightly about three mean directions rather than dispersed widely about one mean direction, as originally thought. It is concluded that the dykes forming the swarm were injected intermittently during several widely distinct periods and that, accordingly, long-range correlation of diabase dyke swarms on the sole basis of their palaeomagnetism may be more hazardous than was realized earlier.

RÉSUMÉ

La publication antérieure de données paléomagnétiques sur le groupe de dykes de diabase précambriens surnommé l'essaim de l'Abitibi y avait indiqué une dispersion angulaire considérable des directions d'aimantations. L'importance relative des différents facteurs auxquels cette dispersion avait été attribuée fait l'objet de la présente étude, laquelle est basée sur un nouvel échantillonnage des mêmes dykes. En grande partie, l'interprétation préliminaire est périmée par suite de cette étude puisque l'aimantation de l'essaim nous apparait non plus comme dispersée autour d'une même direction moyenne mais bien groupée autour de trois directions distinctes. On en tire la conclusion que les dykes formant l'essaim ont été injectés au cours de plusieurs étapes distinctes, ce qui suggère que la corrélation des dykes de diabase à l'échelle continentale et basée seulement sur des données paléomagnétiques comporte plus de dangers qu'on ne l'avait soupçonné au premier abord.

INTRODUCTION

Preliminary results were recently published (Fahrig *et al.* 1965) on the palaeomagnetism of six of the numerous diabase dyke swarms widely distributed through the Canadian Shield. In the analysis of these results, the relatively high angular dispersion noted for the magnetization of some of the swarms was tentatively attributed to a number of causes. To weigh the relative importance of these causes, a detailed study was undertaken on the group of dykes referred to by Fahrig *et al.* (1965) as the "Abitibi swarm". These dykes trend in a general northeasterly direction and they are particularly well exposed in the vicinity of Lake Abitibi and in the Rouyn – Val d'Or area. Both east-northeast- and north-northeast-trending dykes were included in the Abitibi swarm, although a few northerly trending dykes in the area represented in Fig. 1 were considered probably not related to it. Relatively little has been published concerning the age and composition of the Abitibi swarm. In fact, it is possible that the dykes included in it were not all injected at the same time and from a common magmatic source. This is suggested by the 585 m.y. difference between two potassium–argon whole-rock ages (GSC 63-118 and GSC 63-149, Wanless *et al.* 1965) obtained on samples collected from two east-northeasterly trending dykes.

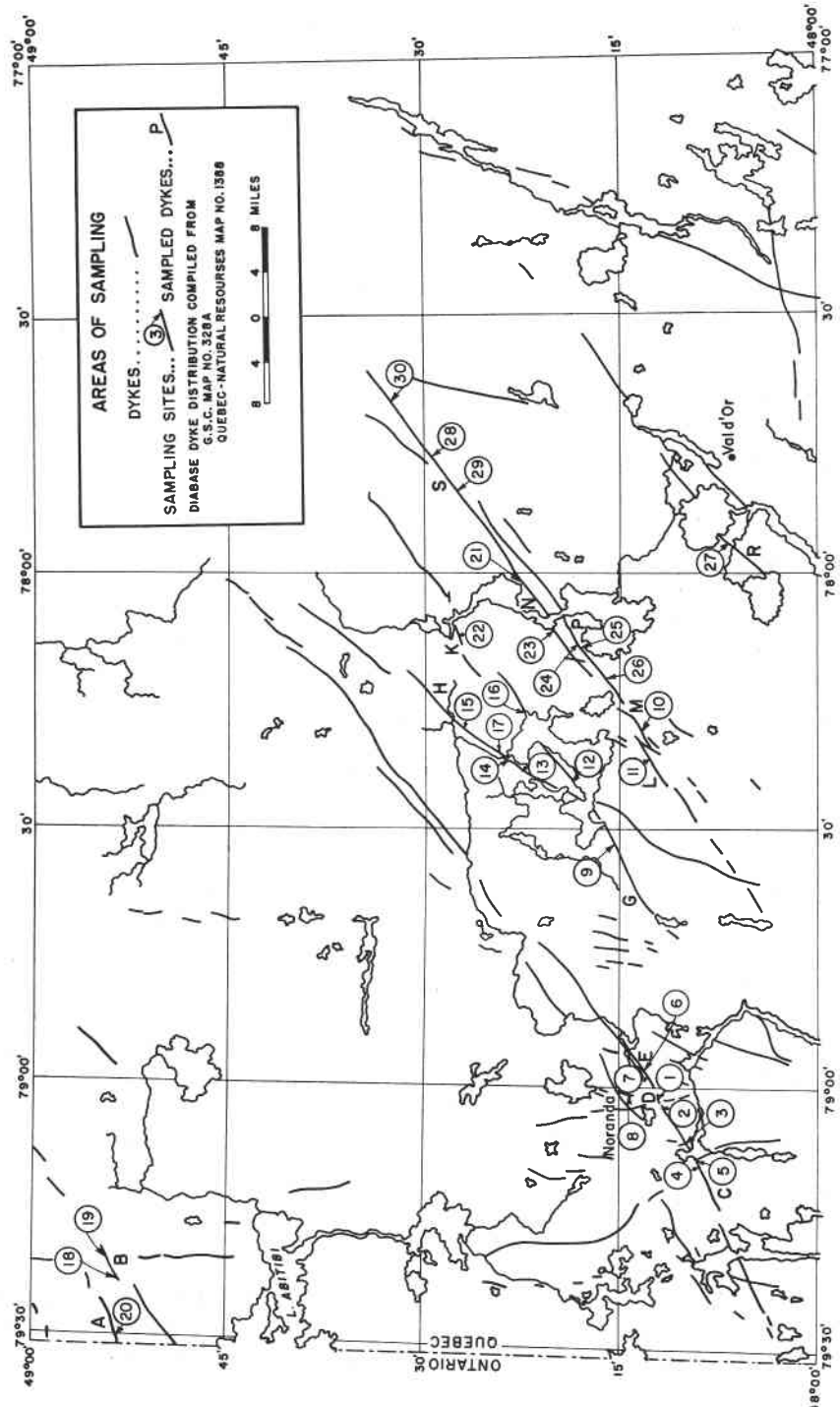


FIG. 1. Distribution of diabase dykes in the Rouyn – Val d'Or area and sampling localities.

The present study indicates that most of the earlier interpretation offered to explain the high angular dispersion in the magnetization of the Abitibi swarm was inappropriate because this magnetization is in fact tightly distributed about three main directions. The data confirm that palaeomagnetism could be useful in correlating diabase dykes within a region, although it appears more hazardous than previously envisaged to correlate swarms located in widely separated areas.

SAMPLING AND MAGNETIZATION MEASUREMENTS

A total of 123 oriented samples were collected from 30 sites (Fig. 1) distributed among 15 individual dykes. With a portable diamond drill, cores (2.54 cm in diameter) were drilled at least 3 meters apart at each site and were oriented *in situ* with a solar compass. From each core, two cylinders, 2.25 cm long, were cut. Each such pair of cylinders will be referred to as a core in the following discussion.

The remanent magnetization of the individual cylinders was measured with a 3-magnet, automatic astatic magnetometer, a description of which will be published later. From the data listed in Table I, the probable errors (standard deviations) in the angular and intensity measurements were calculated as 0.37° and 1.39×10^{-6} c.g.s. units/cc, respectively. These results are based on 10 consecutive measurements of one of the cylinders described above. The cylinder was reoriented in the specimen holder before each of these measurements.

MAGNETIZATION HOMOGENEITY

The two magnetization vectors obtained for each core were normalized to unit length and the direction of their vector sum was taken as the mean magnetization direction of the core. If, however, the two vectors diverged by an angle θ exceeding 23°, the magnetization of the core was not considered sufficiently homogeneous to allow the determination of its direction with reasonable accuracy. In such cases, the mean core directions were neither plotted nor incorporated in the statistical analyses. The tolerance limit for θ was calculated for a minimum within-core precision estimate k of 25 (Fisher 1953). On this basis, the magnetization of more than 93% of the cores examined was found homogeneous (Fig. 2). The natural remanent magnetization of seven cores was inhomogeneous, but after an alternating field treatment in a demagnetizing field of 250 oersteds (peak), six of these cores became homogeneously magnetized (Table II). The natural remanent magnetization intensities of the 123 cores averaged 3.7×10^{-3} c.g.s./cc and they were distributed between 10^{-5} and 10^{-2} c.g.s. units/cc.

ALTERNATING FIELD "CLEANING"

All cores were submitted to a first alternating field "cleaning" treatment with the apparatus described by Larochelle and Black (1965), the amplitude of the demagnetizing field being set at 100 oersteds (peak). A noticeable decrease was noted in the angular dispersion of the magnetizations after remeasurement. The

TABLE I

Results of 10 consecutive magnetization measurements of a cylinder

Declination (deg)	262.8	261.2	262.1	262.7	262.9	261.6	263.1	263.6	263.2	263.2
Inclination (deg)	66.2	66.4	66.3	65.9	66.0	66.0	65.9	66.1	66.3	66.4
Intensity c.g.s./cc $\times 10^{+4}$	2.02867	2.02665	2.02225	2.02798	2.02480	2.02707	2.02525	2.03491	2.03601	2.03111

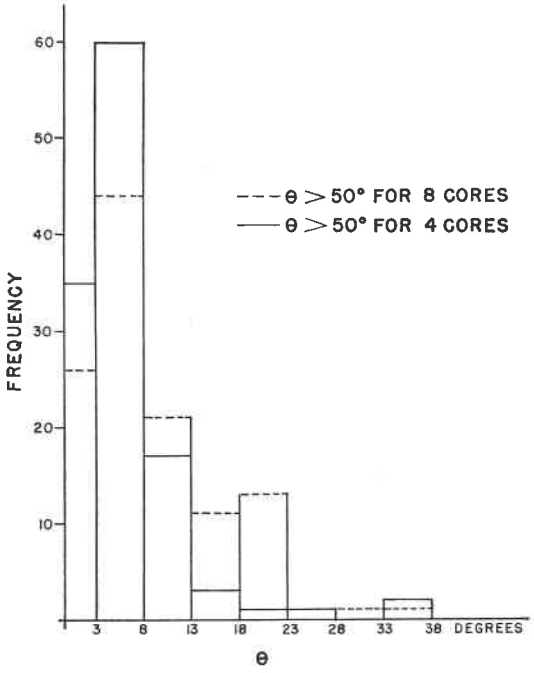


FIG. 2. Histogram of angle θ. Solid horizontal lines refer to data before magnetic cleaning; broken line, to data after cleaning at 250 (100) oersteds (see text).

TABLE II

Natural remanent magnetization (n.r.m.) directions omitted in Figs. 5-A and 5-C

		n.r.m. directions			Direction after 250 oersteds cleaning		
Site No.	Core No.	Decl.	Incl.	θ	Decl.	Incl.	θ
2	3	359.5	−5.3	77.8	289.1	48.3	21.2
5	2	155.5	50.3	99.7	259.5	63.5	4.0
8	4	238.8	16.1	70.1	221.7	5.2	5.1
14	1	175.8	30.0	137.1	301.2	28.4	95.6
19	1	334.3	63.8	34.6	294.6	69.2	19.3
21	4	6.8	34.2	33.5	349.3	−38.7	23.0
29	3	349.4	−5.6	27.3	263.0	56.8	6.4

magnetization of eight cores was no longer homogeneous after this or additional treatments in stronger demagnetizing fields. The data obtained for these cores will not be considered further in this paper.

To investigate if treatments in higher demagnetizing fields would decrease the angular dispersion in the magnetization of the ENE. dykes significantly, one core from each of six different sites was treated successively in demagnetizing fields of 200, 300, 400, 500, 600, and 700 oersteds (peak). Measurements following each treatment (Fig. 3 and Table III) indicated that the magnetization within the group was randomly oriented before the treatment at 200

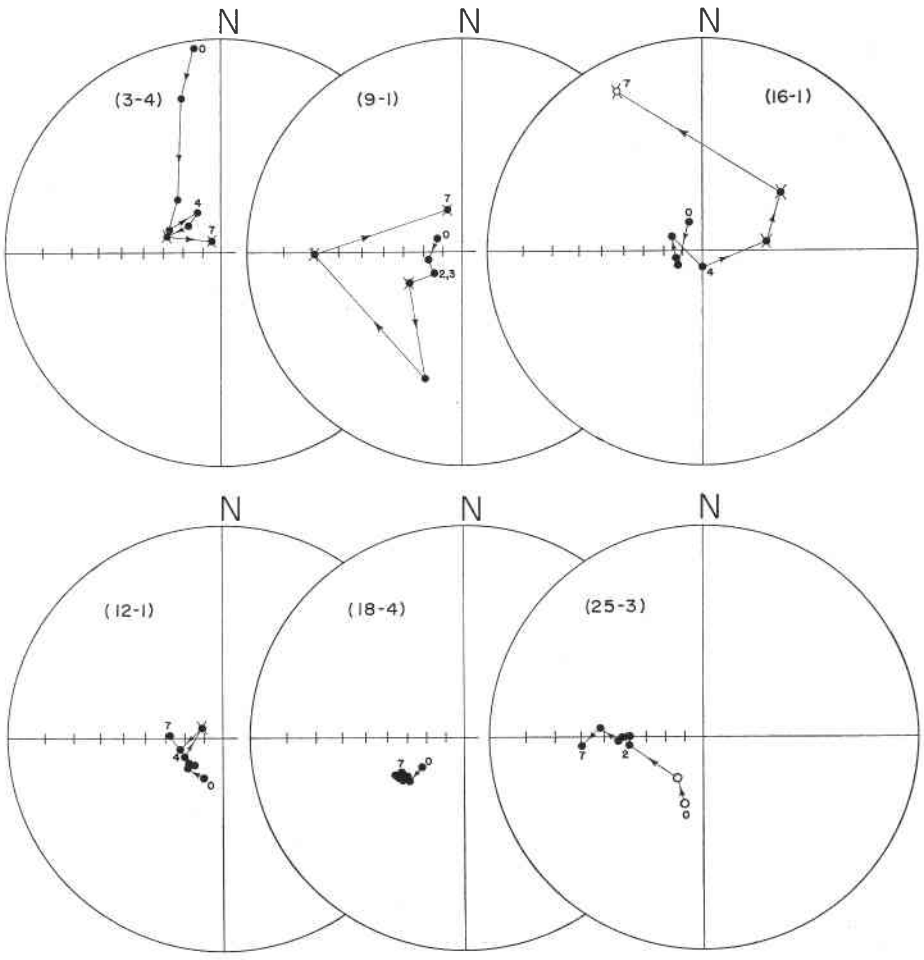


FIG. 3. Magnetization directions of test cores after alternating field treatment under demagnetizing fields whose amplitudes increase by steps of 100 oersteds from 0 to 7×10^2 oersteds. Stereographic projections. Dots crossed with an "X" indicate inhomogeneous magnetization. Open (full) dots indicate upward (downward) pointing magnetization. Numbers in parentheses refer to test-core numbers.

TABLE III

Mean directions of data represented in Fig. 3

Field	Decl.	Incl.	θ_{av}	N	R
0	277.9	68.0	4.8	6	3.172
100	263.2	58.7	5.1	6	3.506
200	257.7	66.0	4.6	6	5.719
300	259.3	65.6	8.3	6	5.798
400	256.6	66.4	10.9	6	5.673
500	240.7	71.1	25.7	6	4.924
600	275.5	58.5	49.4	6	4.858
700	291.6	54.9	46.5	6	4.780

oersteds and that generally the within-core precision began to decrease after the treatment at 300 oersteds, at which point the group dispersion reached a minimum. The mean direction for the group remained practically unchanged after the treatments at 200, 300, and 400 oersteds.

It could be stated intuitively from an examination of Fig. 3 that cores 18-4 and 16-1 are, respectively, the most and least stable of the group. Other indexes, often used in appraising the magnetic stability of rocks, are their isothermal saturation magnetization fields H_{sat}, and the coercivity to their maximum isothermal remanance H_{cr}. These properties were determined for the six cores and the results obtained are plotted in Fig. 4. It is obvious from these

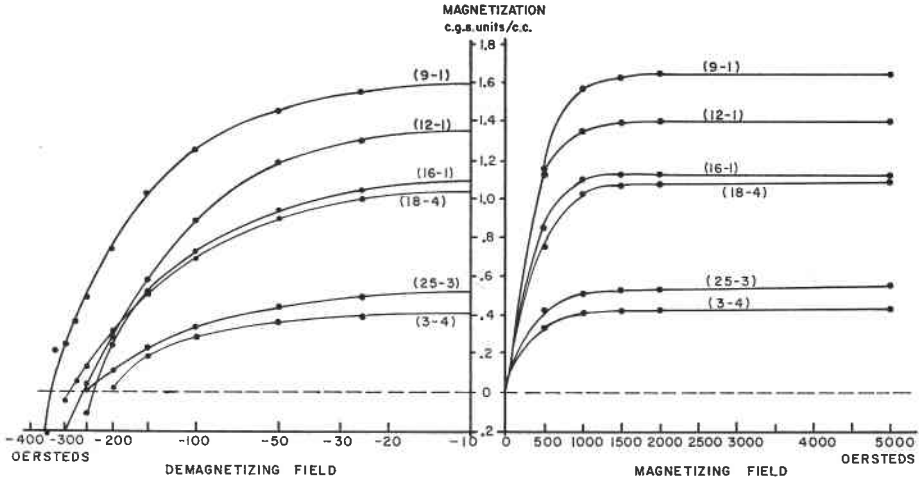


FIG. 4. Isothermal magnetization and demagnetization curves for test cores. Numbers in parentheses refer to test-core numbers.

data that, in the present case at least, there is no simple direct relationship between apparent magnetic stability and either H_{sat} or H_{cr}.

Since some of the cores among the 123 collected could be expected to be less stable than those used in the tests, the peak amplitude for a second general "cleaning" treatment was set at 250 oersteds. After this treatment, five cores were no longer homogeneously magnetized according to the standards set out previously, and their magnetization directions after the first rather than after the second treatment were plotted in Fig. 5 and incorporated in the statistical analyses. It may be verified from Table IV that whether this substitution is made, the overall results are not significantly affected. Only one core exhibited signs of isothermal magnetization due to lightning; its intensity was about 40 times the average after the 250-oersted treatment. Since only four cores were collected from a single site on a north-trending dyke, the results obtained for these cores were not plotted and not interpreted.

Mean directions of magnetization of the treated cores for sites and dykes are

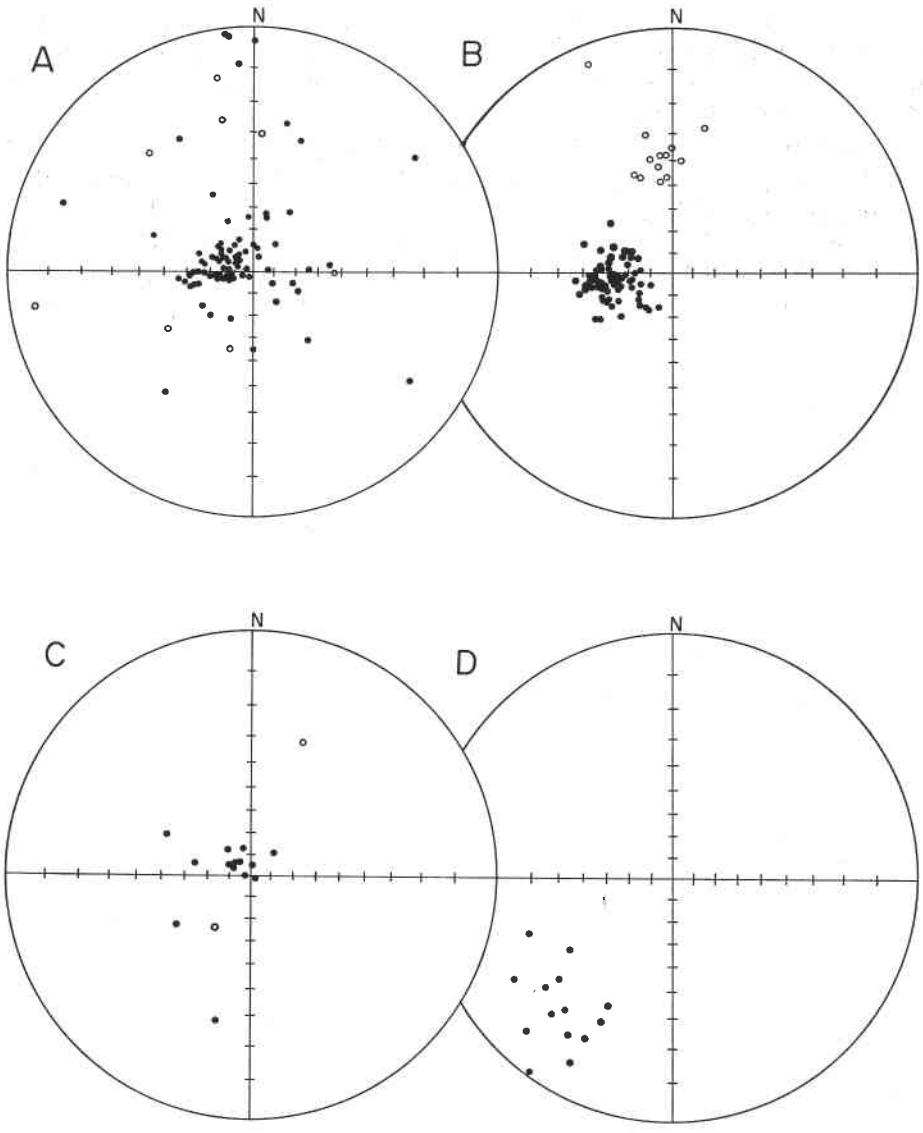


FIG. 5. Stereographic projections of magnetization directions. Open (full) dots indicate upward (downward) pointing vectors. A and C represent natural remanent magnetization directions for cores from ENE. and NNE. dykes, respectively. B and D are respectively for the same groups of data after magnetic "cleaning" at 250 (100) oersteds (see text).

listed separately in Table V for each of the two distinct groups of unit vectors represented in Fig. 5-B and for those in Fig. 5-D. The corresponding values of the precision estimates k (Fisher 1953) for each site and each dyke are also given in the same table along with the same parameter corresponding to the mean core, the mean site, and the mean dyke directions.

TABLE IV

Directions of magnetization of cores no longer homogeneously magnetized after treatment at 250 oersteds

		Direction after cleaning at 100 oersteds			Direction after cleaning at 250 oersteds		
Site No.	Core No.	Decl.	Incl.	θ	Decl.	Incl.	θ
7	3	260.4	70.6	2.6	248.8	73.1	23.6
9	2	269.3	65.0	1.3	238.2	58.1	27.1
9	5	298.6	69.0	9.2	232.3	70.2	56.5
12	4	239.2	72.6	7.6	243.3	57.5	57.5
16	3	270.9	73.3	2.8	197.8	73.8	23.7

STATISTICAL ANALYSIS

The wide range of variation in the value of k either for the 21 sites or the 11 westerly polarized ENE. dykes suggests that the 83 directions of magnetization forming these groups do not belong to populations having the same κ (Watson 1956). The application of another test suggested by Watson (1956) to verify if the various sites or the various dykes have a common mean direction of magnetization also gave negative results. Nevertheless, the results of a double χ^2 test (Table VI) showed at the 95% probability level (*a*) that the 83 directions are distributed symmetrically about their mean, and (*b*) that they diverge from their mean according to Fisher's probability distribution

$$df = \frac{\kappa e^{\kappa \cos \psi} \sin \psi \, d\psi}{2 \sinh \kappa},$$

where κ is the precision index and ψ is the angle between the individual direction and the mean of the group. This illustrates the trivial fact that although a group of vectors may form a Fisherian distribution, they are not necessarily drawn from a single population.

The above results suggest that the westerly polarized ENE. dykes were probably injected intermittently over a relatively long interval of time. On the other hand, the comparatively high within-dyke precision estimates suggest that the dykes did not undergo much differential metamorphism or rotation since the time of their injection. Finally, the high within-site precision estimates suggest that the remagnetization of the dykes by lightning was probably not responsible significantly for the high scattering observed in the preliminary data.

The data on the northerly polarized ENE. dykes are not as revealing because they are less abundant. Nevertheless, there is little if any doubt that dykes M and N were emplaced several million years before or after the other ENE. dykes sampled. A still distinct period of intrusion is indicated for the NNE. dykes, according to the consistent data obtained from the cores collected on dyke H.

DISCUSSION

It is noted first that the mean dyke and mean site directions do not differ significantly within any of the three groups. The mean dyke directions were

TABLE V

Directions of magnetization after "cleaning" in field of 250 oersteds (peak)

Dyke	Site	n	N	k	Decl.	Incl.
D	1	2	2	686	266	53
C	2	3	3	14	257	64
C	3	4	4	91	272	65
C	4	5	5	1840	268	63
C	5	4	4	419	262	65
E	6	4	4	42	279	60
E	7	4	4	573	263	67
G	9	5	5	81	268	72
L	11	4	4	25	262	56
G	12	5	5	89	241	71
ENEW.	—	87	83	55	264	63
ENEW.	—	22	21	98	264	62
A	20	5	5	951	291	68
B	18, 19	8	8	87	248	60
C	2, 3, 4, 5	16	16	77	265	64
D	1	2	2	686	266	53
E	6, 7	8	8	67	272	64
ENEW.	—	11	10	102	265	61
M	10*	5	3	13	356	−23
ENEN.	—	15	13	34	353	−37
ENEN.	—	3	3	56	353	−35
M	10	5	3	13	356	−23
ENEN.	—	2	2	39	354	−32
H	13	5	5	46	211	21
H	15	4	4	27	235	22
NNE.	—	17	14	29	221	19
F	8	4	4	14	219	−3
G	16	3	3	39	258	77
B	18	5	5	275	242	56
B	19	3	3	208	262	66
A	20	5	5	951	291	68
K	22	5	5	430	256	59
P	24	5	5	54	263	59
P	25	5	5	843	271	55
P	26	5	5	146	262	57
R	27*	3	0	—	—	—
S	28	3	3	101	272	56
S	29†	3	2	349	262	54
S	30	2	2	896	277	56
G	9, 12, 16	13	13	66	255	73
K	22	5	5	430	256	59
L	11	4	4	25	262	56
P	24, 25, 26	15	15	113	266	57
R	27*	3	0	—	—	—
S	28, 29, 30†	8	7	195	268	55
N	21	5	5	82	349	−41
N	23	5	5	143	354	−41
N	21, 23	10	10	110	352	−41
H	14*	3	1	—	213	15
H	17*	5	4	55	221	15
NNE.	13, 14, 15, 17	4	4	53	220	18

NOTE: n is the number of cores collected; N is the number of cores incorporated in statistics.
*$(n - N)$ cores rejected because of magnetic instability.
†One core rejected because of isothermal magnetization by lighting.

TABLE VI

Results of χ^2 test to verify if ENEW. group has Fisher distribution

																			$\Sigma (O-E)^2/E$	$\chi^2_{.05}$
(*a*)	L	22.5	45	67.5	90	112.5	135	157.5	180	202.5	225	247.5	270	292.5	315	337.5	360			
	E	5.2	5.2	5.2	5.2	5.2	5.2	5.2	5.2	5.2	5.2	5.2	5.2	5.2	5.2	5.2	5.2			
	O	4	5	6	1	4	6	5	6	8	5	6	8	2	6	5	6		9.7	25.0
(*b*)	L	3	5	6	7	8	9	10	11	12	14	16	180							
	E	6	9.6	5.9	6.3	6.5	6.4	6.2	5.8	5.3	8.8	6.4	10.0							
	O	14	8	4	5	5	6	9	3	4	8	7	10						15.3	18.3

NOTE: L = upper limits of intervals in degrees.
E = expected number within interval.
O = observed number within interval.

used to calculate the palaeomagnetic pole positions in the three instances (Table VII).

The necessity of grouping the magnetization directions according to dyke trends and, within the ENE. dykes, according to polarity, appears obvious.

The northerly pointing vectors plotted in Fig. 5-B are those obtained for the cores collected at sites 10, 21, and 23, which are distributed over a distance of 30 km along dykes M and N. The consistency of the data considerably reduces the chance that these samples were magnetized isothermally by lightning, after their initial cooling.

TABLE VII

Pole positions calculated from mean dyke directions

	N	R	k	α_{95} (deg)	Decl. (deg)	Incl. (deg)	Long. (deg)	Lat. (deg)	δm (deg)	δp (deg)
ENEW.	10	9.912	102	4.8	264	61	134 W.	27 N.	7.4	5.7
ENEN.	2	1.974	39	—	354	−32	107 E.	24 N.	—	—
NNE.	1	1	—	—	221	19	122 W.	21 S.	—	—

The mean direction of magnetization obtained for the westerly polarized ENE. dykes is very close to that obtained for the Abitibi swarm in the preliminary study (Fahrig *et al.* 1965), although the high angular dispersion of the preliminary data had made their reliability questionable. This high dispersion could be attributed (1) to orientation sampling errors, (2) to the fact that the directions of magnetization of the Abitibi swarm are formed by a number of distinct although non-random populations, or (3) to anhysteritic magnetization introduced in some samples during the original "cleaning" treatment.

To test this last possibility, the Abitibi dyke samples discussed in the preliminary study (Fahrig *et al.* 1965) were resubmitted to alternating field "cleaning" treatments with the apparatus mentioned earlier. The peak amplitude of the demagnetizing field was set above that of the last cleaning treatment to which the individual sample had been submitted. The directions of magnetization of the samples previously accepted as stable did not vary significantly as a result of this treatment. With two exceptions only, the samples which had previously been considered unstable in the preliminary study were found homogeneously magnetized. None of the samples displayed an intensity of magnetization indicative of isothermal magnetization due to lightning and the only two samples which had been collected from dyke M were also polarized northerly upward. This last result is interpreted as another indication that the ENE. dykes in the area considered were injected during at least two widely separated geological periods. However, much scattering remained in the data, even after they were grouped according to dyke trends and polarity (for example, $k = 14$, for the ENE. westerly polarized dykes), and it is suggested that most of it is related to orientation sampling errors, since the same high scattering is not present in the data described earlier in the present study.

CONCLUSIONS

(1) Since the within-dyke dispersion is relatively low for the data described in this paper, it is suggested that lightning, differential metamorphism, and (or) rotation are not valid explanations for the angular dispersion observed in the preliminary data. The "random populations" postulated earlier appear to be the result of sampling orientation errors and of inadequate sampling at each site. More abundant sampling from dyke M, for instance, would have allowed us to detect that the magnetization of this dyke belonged to a distinct although non-random population.

(2) Palaeomagnetism appears to be a reliable tool for geological correlation of diabase dykes within a limited region.

(3) The group of dykes discussed in the present study appears to have been injected during more than one interval of the Precambrian, and any one of these intervals may have extended over several million years.

(4) For the above reasons, long-range correlation of diabase dyke swarms on the sole basis of their palaeomagnetism may be more hazardous than realized previously, although the potential of this tool should not be discarded without further study.

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